Market Making on Binance Perpetual Futures

Progress Report: Data Preparation, Volatility, Tails, Adverse Selection and Fees

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Summary

This report covers the first three stages of the project: building and checking the data (Step 0), testing whether volatility is constant and returns are normal (Q5 and Q6), and measuring what market makers gain and lose on each trade (Q1–Q3). Everything here uses January and February 2024 only. March 2024 is held back, untouched, for a final out-of-sample check.

- **The data are sound.** All 90 days are complete for all three contracts. The only defects are three short gaps in Binance's quote archive, about 0.1% of the sample, which are excluded. Trades and quotes line up almost perfectly: 99.7–99.9% of trades fall on the side of the market their direction flag implies.
- **Volatility is far from constant.** It is about twice as high at 15:00 UTC as in the quietest hour, 20–40% lower at weekends, and it comes in bursts that take 40–50 minutes to fade. A market maker who assumes constant volatility, as the Avellaneda–Stoikov model does, misprices quotes by more than one tick in the most volatile 5% of minutes.
- **Returns are far from normal.** Most of the extreme moves are explained by changing volatility, but even after accounting for it, moves beyond four standard deviations are 33–48 times more common than a normal distribution predicts.
- **Being filled is bad news for a passive order.** Within one second of a trade, the price has already moved against the market maker by more than the spread they earned. On average a resting order loses money before fees on all three contracts, so at standard fee levels passive market making only pays for participants who receive fee rebates.

Next: the fill-probability model, the secondary questions, and evaluating every fitted model on March.

1 Introduction

1.1 The question

A *market maker* continuously offers to buy (a *bid*) and to sell (an *ask*) and earns the gap between the two, the *spread*. The best-known model of how a market maker should set these prices is that of Avellaneda and Stoikov (2008) (A–S), which also underlies many trading simulators, including the EvoMM simulator used in Prof. O'Connor's group. The model makes three simplifying assumptions:

1. prices move randomly with **constant volatility** and **normally distributed** changes;
2. the chance that a quote is traded against **falls off exponentially** with its distance from the current price;
3. being traded against **carries no information** about where the price goes next.

The project tests these assumptions on real data (Part A) and measures what market making actually earns (Part B). It has two clients of equal priority: **research groups that build market-making simulators**, who need to know which assumptions to relax and what realistic parameter values look like, and a **crypto market-making desk**, which needs to know which contracts are worth quoting, when, and whether it can cover fees.

1.2 Data

Best bid/ask updates and individual trades from Binance's public archive (Binance, 2024) for three perpetual futures contracts, ETHUSDT, LINKUSDT and MATICUSDT, from 1 January to 30 March 2024 (90 days). The three contracts range from very heavily traded (ETH, about $6 billion a day) to much less so (LINK and MATIC, about $300–400 million a day).

1.3 Key terms

Table 1 defines the terms used throughout.

Term	Meaning
Mid price	Halfway between the best bid and the best ask; the "current price"
Tick	The smallest price step the exchange allows ($0.01 for ETH, $0.001 for LINK, $0.0001 for MATIC)
Basis point (bps)	One hundredth of one percent. 1 bps on a $3,000 price is $0.30
Taker / maker	The trader who accepts a waiting price / the trader whose waiting order was accepted
Parent trade	All the individual fills one order produces in the same millisecond, counted as one trade
Half-spread	What a maker earns per fill relative to the mid price: half the gap between bid and ask
Markout	How far the mid price moves in the taker's direction in the seconds after a trade; the maker's loss to better-informed traders (*adverse selection*)
Realized half-spread	What the maker actually keeps after the price has moved: half-spread minus markout
Realized volatility	Volatility measured from observed price changes within an interval
Volatility regime	Whether the past 30 minutes were calm, normal or turbulent (lowest, middle or highest third)

Table 1: Terms used in this report.

2 Methodology and Results

This section takes each completed stage in turn. Each part first says what was done and which statistical method was used, then what the data showed. All estimates use January–February 2024; the 95% intervals account for dependence between nearby observations, either with Newey–West standard errors or by resampling whole days.

2.1 Step 0: Preparing and checking the data

2.1.1 What was done

The raw files contain about 2.8 billion quote updates and 500 million trades. A processing script reads each contract-day once, checks it, cleans it, and writes three compact tables that every later analysis uses:

- a **one-second price table**: bid, ask and mid price at the end of every second (86,400 rows a day);
- a **one-minute table**: volatility, returns, trading volume, buying and selling pressure, spread and the size available at the best prices (1,440 rows a day);
- a **parent-trade table**: one row per order, with its direction, size, the prices just before it and the mid price 1 to 60 seconds afterwards.

Five design choices matter for everything that follows:

- Prices are recorded on a **fixed clock** (every second or minute), because quote updates arrive at irregular times and the A–S model is written in clock time.

- Averages over quotes are **weighted by how long each quote lasted**, so that a few hectic seconds with hundreds of updates do not outweigh long calm stretches.
- Each trade is matched to the **last quote strictly before it**, using the exchange's matching-engine clock. A quote stamped in the same millisecond is usually the order book *after* the trade hit it.
- Individual fills from one order are merged into **parent trades**, so that one large order is not counted as ten separate trades. On average about five ETH fills make one parent trade.
- Any stretch longer than 60 seconds without a quote update is treated as **missing data** and excluded, but whole days are never dropped.

2.1.2 What the checks showed

	ETH	LINK	MATIC
Days complete	90/90	90/90	90/90
Quote updates (millions)	2,199	285	310
Trades (millions)	350	80	71
Duplicate, crossed or zero-size quotes	0	0	0
Days with out-of-order rows (re-sorted)	23	0	0
Missing trades	0.009%	0.020%	0.024%
Trades on the correct side of the quote (days without gaps)	99.74%	99.81%	99.90%
Time the spread sat at exactly one tick (Jan–Feb)	99.6%	99.5%	99.8%
Half-spread earned per fill (Jan–Feb)	0.021 bps	0.307 bps	0.586 bps

Table 2: Data-quality checks. "Correct side" means a buyer-initiated trade printed at or above the ask, or a seller-initiated trade at or below the bid.

- **Archive gaps.** Three gaps appear at the same times on all three contracts: 3 January (12:10 UTC, about 13 minutes), 28 February (17:33 UTC, about 43 minutes) and 8 March (15:35 and 16:04 UTC, about 86 minutes in total). Trades kept being recorded during them, so these are holes in Binance's quote archive, not trading halts. They total about 144 minutes per contract, roughly 0.1% of the sample, and are excluded from every quote-based analysis.
- **Trade direction is reliable.** Away from the gap days, 99.7–99.9% of trades fall on the expected side of the quote, and the best alignment occurs with no time offset at all. At most 0.6% of trades could be misclassified, which would shrink the markouts in Step 2 by about 1%.
- **Every spread is pinned at one tick.** All three contracts spend more than 99% of the time at the smallest spread the exchange allows. What differs between them is how much a tick is worth: about 0.04 bps of the price for ETH, 0.6 bps for LINK and 1.2 bps for MATIC. Liquidity therefore varies mainly through the size waiting at the best prices, not through the spread.

2.2 Q1: Is volatility constant?

2.2.1 Methods

- **Time-of-day pattern:** a regression of each minute's log realized volatility on 24 hour-of-day indicators and a weekend indicator, with Newey–West standard errors (Newey and West, 1987), which allow for correlation between nearby minutes.
- **Clustering:** autocorrelation of returns and of absolute returns, and the HAR regression of Corsi (2009), which predicts next-minute volatility from the last minute, the last 30 minutes and the last day.
- **GARCH models** (Bollerslev, 1986; Glosten et al., 1993) with normal and Student-t errors, fitted after removing the time-of-day pattern. These give the *half-life* of a volatility shock: how long until half of a burst has faded.
- **Materiality:** in the A–S model, how far the quote is shifted to manage inventory is proportional to the variance. If a desk uses the average variance when the true variance is v times the average, its quote is wrong by $X(v-1)$ ticks, where X is its usual inventory adjustment in ticks. The proposal fixed in advance that an error above one tick counts as a material violation.

2.2.2 Results

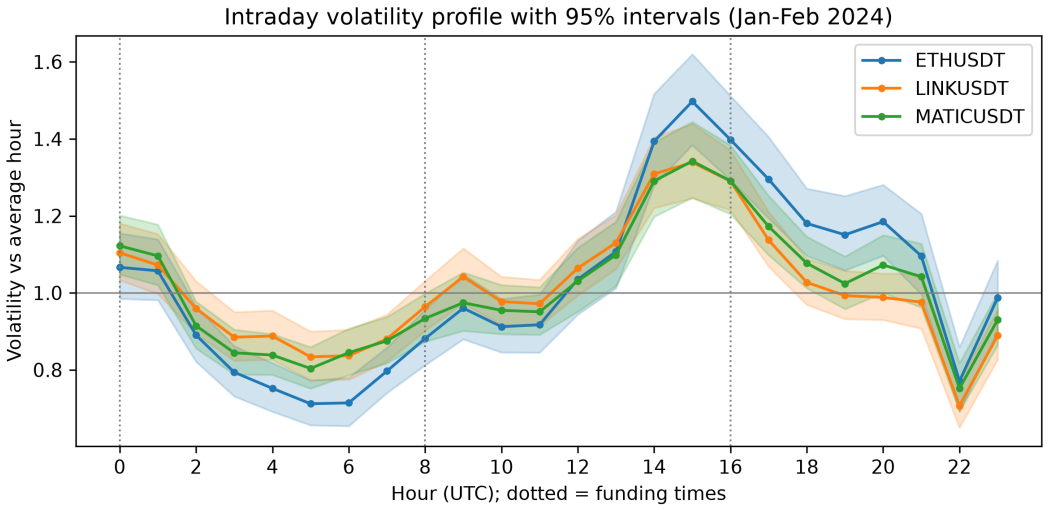


Figure 1: Volatility by hour of the day relative to the average hour, with 95% intervals. All three contracts peak at 15:00 UTC, around the US market open. Dotted lines mark the 8-hourly funding times.

Time of day. Volatility follows a strong and shared daily rhythm (Figure 1). It is quietest in the early UTC morning and at 22:00 UTC and busiest between 14:00 and 17:00 UTC.

	ETH	LINK	MATIC
Busiest / quietest hour (UTC)	15 / 05	15 / 22	15 / 22
Busiest vs quietest hour [95% CI]	2.10× [1.87, 2.36]	1.89× [1.69, 2.12]	1.78× [1.59, 2.00]
Weekend vs weekday [95% CI]	0.62× [0.59, 0.65]	0.79× [0.76, 0.83]	0.77× [0.74, 0.80]
Same ratio without the ETF-approval days	2.08×	1.88×	1.78×

Table 3: Time-of-day and weekend effects on volatility. Because variance is volatility squared, the busiest hour has 3.2–4.4 times the variance of the quietest.

There is also a sharp one-hour lull at 22:00 UTC, about 20–25% below the neighbouring hours. It appears on 87–93% of days and on every day of the week, including Saturdays. That rules out the explanation I first considered, the daily pause in CME's crypto futures, because CME is closed on Saturdays. The cause is still open.

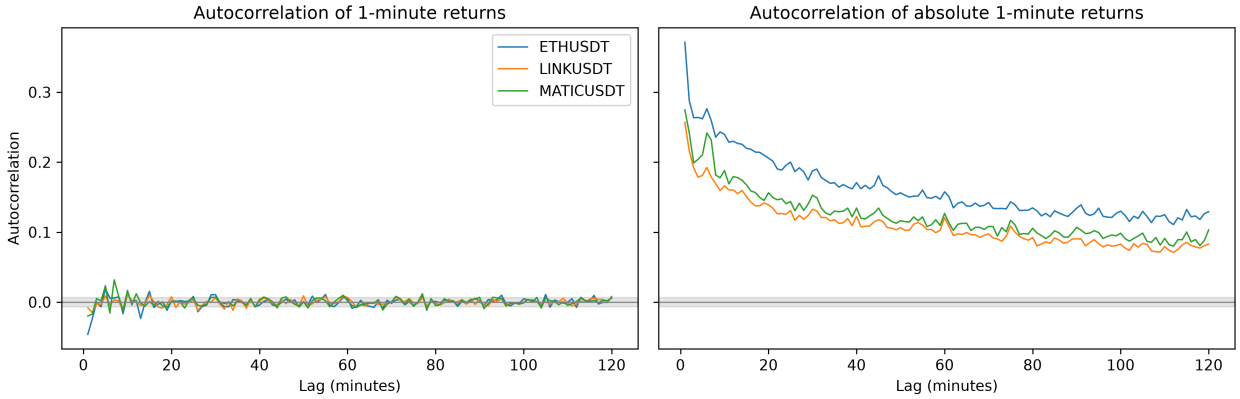


Figure 2: Autocorrelation of one-minute returns (left) and of their absolute size (right). Grey bands show the range expected by chance. The direction of the next move is almost unpredictable, but its size is strongly predictable for hours.

Clustering. Returns themselves are almost unpredictable from one minute to the next (autocorrelations of at most 0.05 in size), but their *size* is not (Figure 2). A large move is followed by more large moves: the correlation is 0.26–0.37 with the previous minute and still 0.12–0.16 an hour later. In the HAR regression, the past 30 minutes is the best guide to the next minute (coefficient 0.55–0.58), ahead of the last minute (0.32–0.36) and the last day (0.08–0.10), and the model explains about 60% of minute-to-minute variation in volatility.

Two layers of volatility. A standard GARCH model with Student-t errors fits far better than one with normal errors, but for ETH it reports volatility shocks that never fade (persistence exactly 1). Fitting January and February separately does not change this. Dividing out each day's own volatility level does:

Half-life of a volatility shock	ETH	LINK	MATIC
GARCH with Student-t errors	not identifiable	98 min	174 min
Same, after removing each day's level	**37 min**	**38 min**	**50 min**
Student-t degrees of freedom	6.3	7.2	6.9

Table 4: Persistence of volatility. The second row isolates bursts within a day.

Volatility therefore moves on two timescales: bursts within a day that halve in about 40–50 minutes (strikingly similar across the three contracts), on top of a daily level that drifts slowly and, for ETH, almost permanently over these two months. A single GARCH model mixes the two and overstates how long bursts last.

Is the violation material? Figure 3 shows how far each hour's variance is from the single constant value the A–S model assumes.

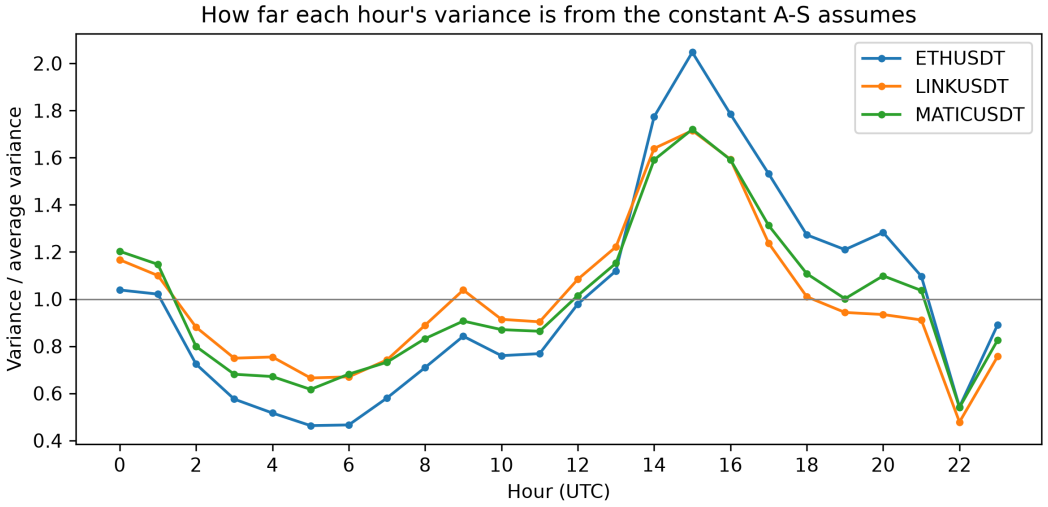


Figure 3: Each hour's variance divided by the average variance. A constant-volatility model assumes this line is flat at 1.

Quote error for a desk whose usual adjustment is 1 tick	ETH	LINK	MATIC
At 15:00 UTC, from the time of day alone	**1.05 ticks**	0.71	0.72
In the most volatile 5% of minutes	**1.90**	**1.49**	**1.51**
In the most volatile 1% of minutes	**6.2**	**4.3**	**4.7**

Table 5: Error from assuming constant volatility, in ticks. The error scales in proportion to the desk's adjustment; values above one tick (bold) meet the materiality threshold set in the proposal.

By the threshold set in the proposal, the constant-volatility assumption is materially violated on all three contracts in volatile periods, and on ETH from the time of day alone.

2.3 Q2: Are price changes normally distributed?

2.3.1 Methods

For one-second, ten-second and one-minute returns: the share of intervals with no price change, skewness, excess kurtosis (zero for a normal distribution), the share of moves beyond four standard deviations, a Student-t fit compared with a normal fit, and the Hill estimator of how heavy the tail is (Hill, 1975). The key step is to repeat everything after dividing each return by its GARCH volatility: if what remains looks normal, the extreme moves came from changing volatility; if not, there are genuine jumps as well.

2.3.2 Results

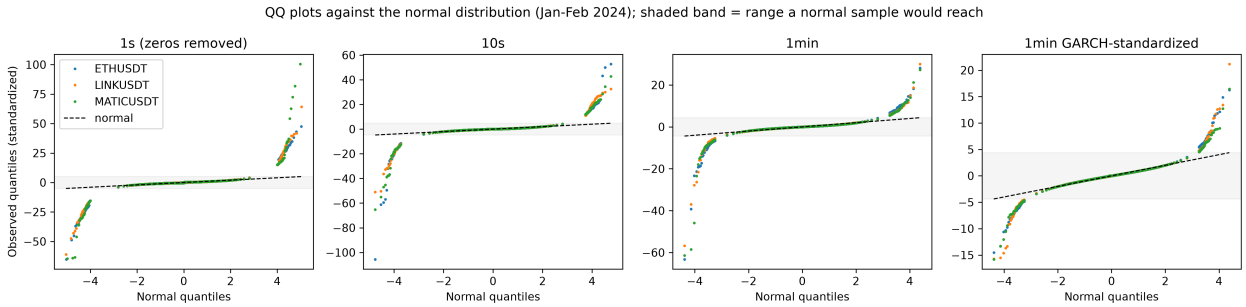


Figure 4: Observed returns against a normal distribution. Points on the dashed line would mean normality. Points far outside the shaded band are moves a normal distribution would essentially never produce. The right-hand panel shows one-minute returns after accounting for changing volatility.

One-minute returns	ETH	LINK	MATIC
Excess kurtosis: raw	243	175	372
after removing the time-of-day pattern	48	52	68
after also accounting for clustering (GARCH)	6.4	8.6	4.9
Moves beyond 4 SD, relative to a normal: raw	87×	59×	67×
after GARCH	48×	33×	40×
Largest move after GARCH (standard deviations)	16	21	16
Student-t degrees of freedom after GARCH	6.3	7.2	7.0
One-second intervals with no price change	62%	58%	70%

Table 6: How far returns are from normal, before and after accounting for changing volatility.

- Time of day plus volatility clustering explain about 97% of the excess kurtosis. The large negative skew in raw returns (as low as −6) also disappears, so it came from a few crash minutes during volatile spells rather than a built-in downward bias.
- What remains is still not normal (Figure 4, right): moves beyond four standard deviations are 33–48 times more common than a normal distribution predicts, and the largest standardized moves reach 16–21 standard deviations. There are genuine jumps on top of changing volatility.
- At the one-second scale the mid price does not move at all in 58–70% of seconds, so a smoothly moving random walk is a poor description of prices at the timescale where market makers operate.

2.4 Q4: Do fills carry information?

2.4.1 Methods

For each of the 67 million parent trades in January and February, the *markout* at horizon h is

$$\text{markout}(h) = s \cdot \frac{m_{t+h} - m_t}{m_t} \times 10^4 \text{ bps},$$

where $s = +1$ for a buyer-initiated trade and -1 for a seller-initiated one, and m_t is the mid price just before the trade. A positive markout means the price kept moving the way the taker wanted, which is a loss for the maker. Averages come with 95% intervals from a *day-block bootstrap* (Künsch, 1989), which resamples whole days and so respects the dependence between trades on the same day. Trades during archive gaps, or matched to a quote more than five seconds old, are excluded (about 1%).

2.4.2 Results

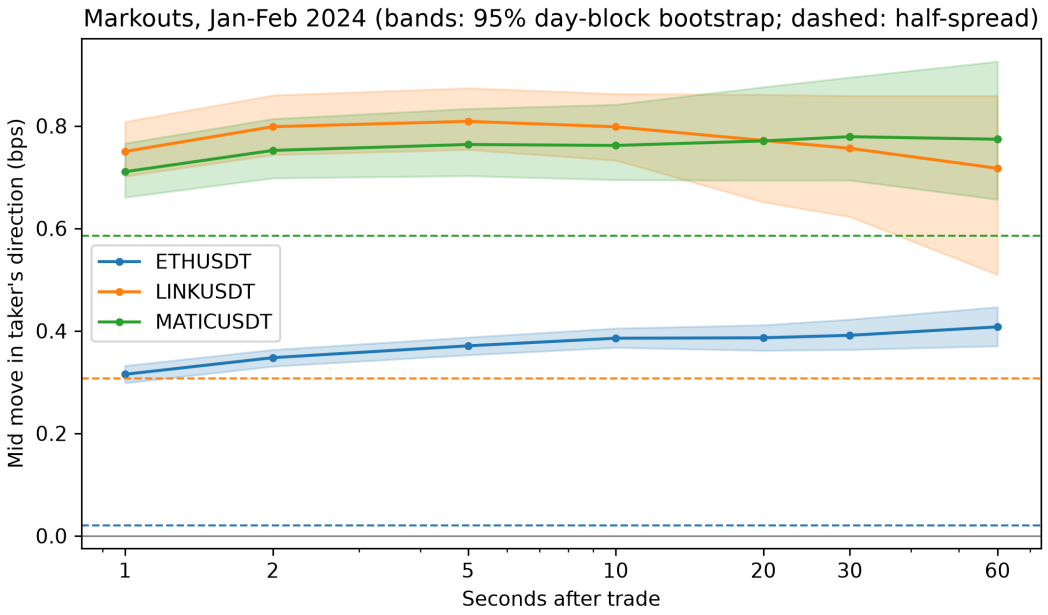


Figure 5: Average price move in the taker's direction after a trade, with 95% day-block bootstrap intervals. Dashed lines show each contract's half-spread, the amount a maker earns per fill. Curves above their dashed line mean the average maker loses money before fees.

bps, [95% CI]	Half-spread earned	After 1 s	After 5 s	After 60 s
ETH	0.02	0.32 [0.30, 0.33]	0.37 [0.35, 0.39]	0.41 [0.37, 0.45]
LINK	0.31	0.75 [0.70, 0.81]	0.81 [0.75, 0.87]	0.72 [0.51, 0.86]
MATIC	0.59	0.71 [0.66, 0.77]	0.76 [0.70, 0.83]	0.77 [0.66, 0.93]

Table 7: Markouts compared with the half-spread a maker earns.

- The A–S assumption that fills carry no information fails clearly. Within one second the price

has moved against the maker by more than the half-spread on every contract, and on ETH by about fifteen times the half-spread (Figure 5).

- Most of the move happens almost immediately: at least three-quarters of the 60-second markout is already there after one second. On ETH the price keeps drifting the same way, which points to lasting information. On LINK it peaks after about five seconds and partly reverses, suggesting part of the move is temporary pressure.
- Leaving out the ETF-approval days changes these numbers only in the second decimal place.

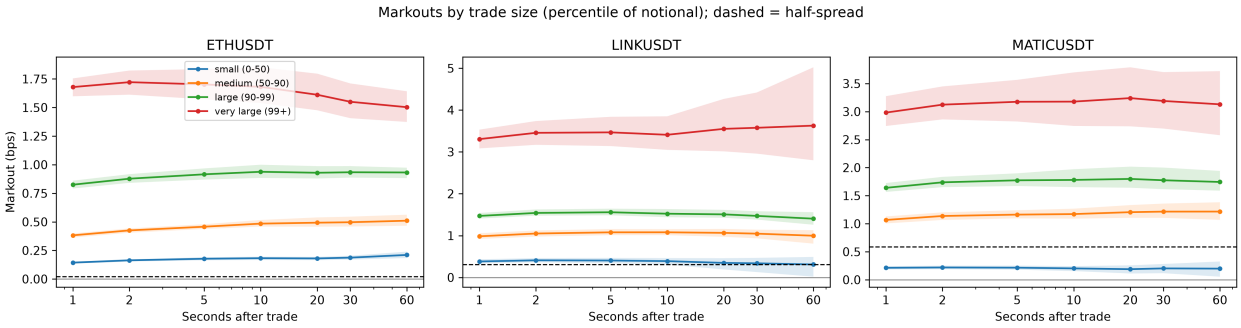


Figure 6: Markouts by trade size, measured as percentiles of trade value within each contract. Larger trades move the price much more.

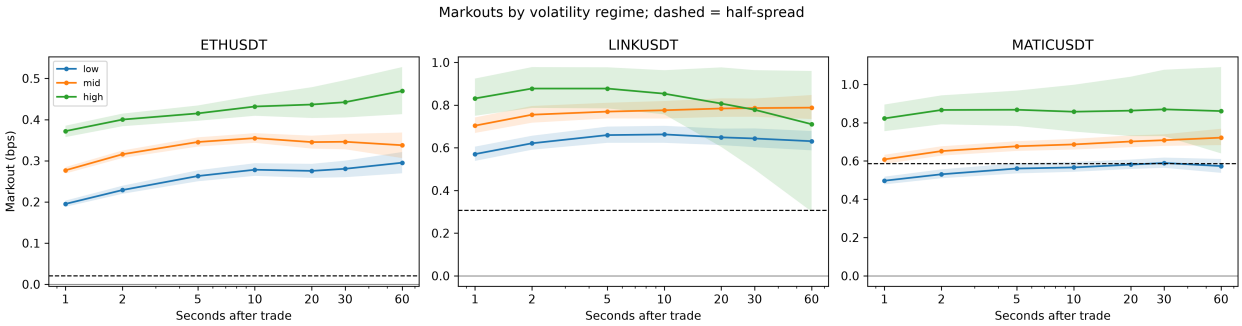


Figure 7: Markouts by volatility regime (how turbulent the previous 30 minutes were). Adverse selection is higher when markets are volatile.

Size and volatility. The markout rises steeply with trade size (Figure 6): after 60 seconds it is 0.2–0.3 bps for the smallest half of trades and 1.5–3.6 bps for the largest 1%. Larger orders are more likely to come from traders who know something. Adverse selection is also 50–60% higher in the most volatile regime than in the calmest on ETH and MATIC (Figure 7); on LINK the difference is smaller and noisier.

Does order flow predict the next minute? Barely. A regression of the next minute's return on this minute's buying-versus-selling pressure and on the balance of size waiting at the bid and ask explains under 0.3% of the variation, and a logistic model of the next minute's direction has an AUC of 0.51–0.52, hardly better than a coin flip. The balance of waiting size (*quote imbalance*) is the more useful of the two signals. Taken together with the markouts, this means the information in a trade is absorbed within seconds and is mostly gone by the next minute.

2.5 Q5: What do market makers actually keep?

2.5.1 Methods

Each trade's cost to the taker splits into two parts (Huang and Stoll, 1996; Hasbrouck, 2007):

$$\underbrace{s\,\frac{p - m_t}{m_t}}_{\text{effective half-spread: earned at the fill}} = \underbrace{s\,\frac{p - m_{t+h}}{m_t}}_{\text{realized half-spread: kept}} + \underbrace{s\,\frac{m_{t+h} - m_t}{m_t}}_{\text{price impact: lost}},$$

where p is the trade's average price. The realized half-spread is what a maker keeps per fill after the price has moved, before paying any fee. A regression of the realized half-spread on trade size, time of day and volatility regime, with errors clustered by day, shows what drives it.

2.5.2 Results

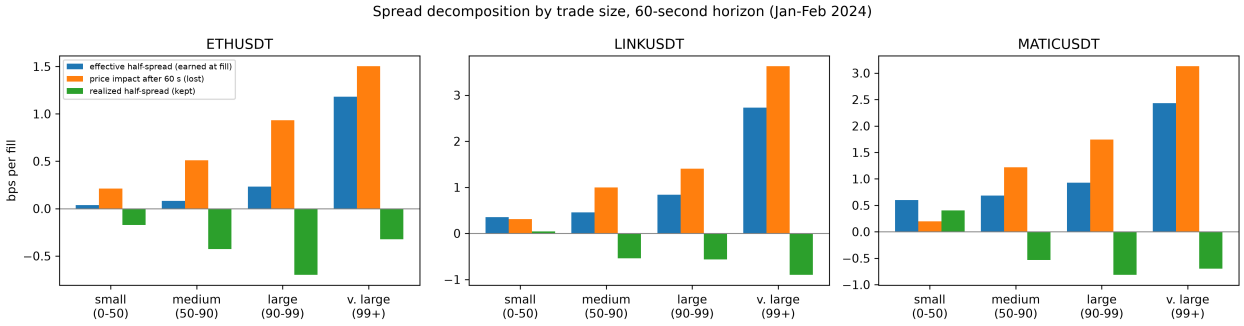


Figure 8: What a maker earns at the fill (blue), loses to the following price move (orange) and keeps (green), by trade size, 60 seconds after the trade.

bps per fill, before fees	Earned	Lost (60 s)	Kept (5 s)	Kept (60 s)	Kept, small trades
ETH	0.08	0.41	−0.29	**−0.32**	−0.17
LINK	0.46	0.72	−0.35	**−0.26**	+0.04
MATIC	0.68	0.77	−0.08	**−0.09**	+0.40

Table 8: Spread decomposition per fill. "Small trades" are the smallest half of trades by value.

- On average a maker keeps a *negative* amount on every contract: the price move after a fill outweighs what the fill earned. MATIC's 95% interval, $[-0.20, 0.02]$ bps, just reaches zero. The result is the same at five seconds, so it does not depend on how long a maker holds the position.
- The losses come from larger trades (Figure 8). Only small trades on MATIC and LINK leave the maker with something. In the regression, medium and large trades cost the maker 0.2–1.3 bps more per fill than small ones; time of day has no significant effect.
- The maker earns more at the fill than the quoted half-spread (0.08 against 0.02 bps on ETH) because larger orders sweep past the best price and pay more.

2.6 Q6: Can a small market maker cover fees?

2.6.1 Methods

The net gain per fill is the realized half-spread at 60 seconds minus the maker fee, evaluated over a range of fees from a 0.5 bps rebate to a 2 bps charge. The *break-even fee* is the fee at which the average net gain is zero. A logistic regression of whether a fill is profitable at a 2 bps fee on trade size, hour of day and volatility regime, with errors clustered by day, tests whether simple observable conditions can pick out the good fills; its AUC measures how well it separates them (0.5 is a coin flip, 1 is perfect).

2.6.2 Results

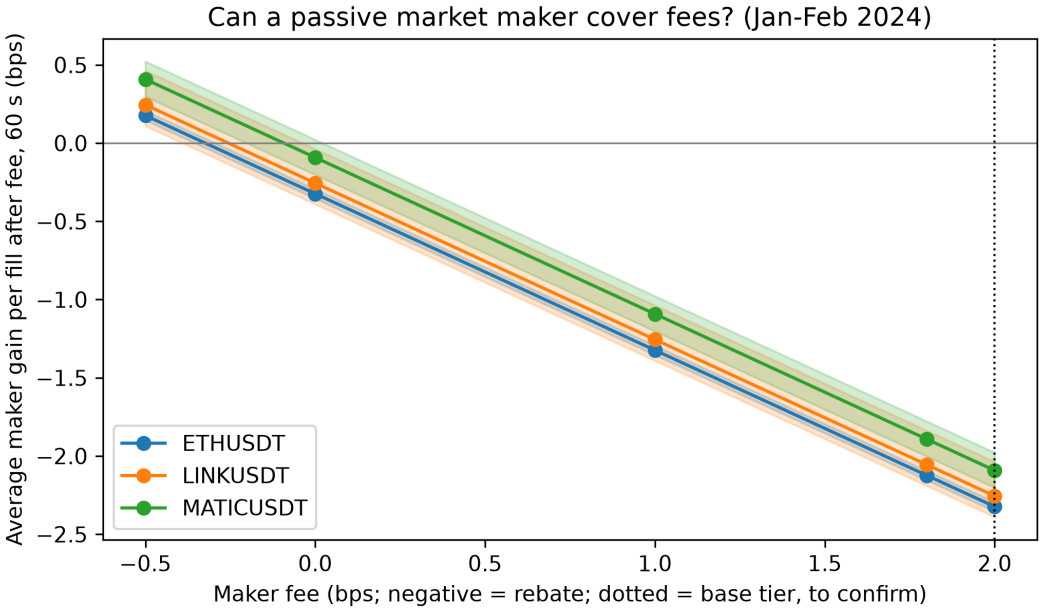


Figure 9: Average gain per fill after paying the maker fee, against the fee level. Negative fees are rebates. The line crosses zero at the break-even fee. The dotted line marks the base-tier fee of about 2 bps

	Break-even maker fee	Net per fill at a 2 bps fee	Fills profitable at 2 bps
ETH	−0.32 bps	−2.32 bps	37%
LINK	−0.26 bps	−2.26 bps	42%
MATIC	−0.09 bps	−2.09 bps	41%

Table 9: Fees and passive-fill profitability, 60 seconds after the fill.

- The break-even fee is **negative** on all three contracts (Figure 9): on average, providing liquidity only pays if the exchange pays a rebate for it. This holds whatever the exact standard fee turns out to be.
- About 40% of individual fills still clear a 2 bps fee because prices often reverse, but the losing fills lose more than the winning ones gain.

- Simple rules cannot pick the good fills. A logistic model using trade size, time of day and volatility regime has an AUC of only 0.53–0.57. High volatility raises the chance that a single fill clears the fee, but it raises the chance of a large loss just as much, so the average does not improve.

2.7 What the results so far mean for each client

Research groups building simulators.

- **Constant volatility** should be replaced by three components: a time-of-day profile (peak 1.8–2.1 times the quietest hour, weekends 0.6–0.8 times weekdays), a two-layer volatility process (a slowly drifting daily level times intraday bursts with a 40–50 minute half-life), and Student-t price shocks with about 6–7 degrees of freedom. Even this underpredicts the most extreme minutes, so a jump component is the next refinement.
- **No information in fills** is the most wrong of the assumptions tested so far. Adverse selection should act within seconds, grow steeply with trade size, rise in volatile periods, and be large enough that the average passive fill loses money. Without it, strategies tuned in simulation will look far more profitable than they can be in practice.

Market-making desk.

- At standard fees the average passive fill loses money on all three contracts; quoting them is only viable with maker rebates.
- The least costly flow is small trades on MATIC and LINK. Being filled by large orders is where the losses are.
- Risk is highest from 14:00 to 17:00 UTC and lowest in the early UTC morning, at 22:00 UTC and at weekends. The volatility of the past 30 minutes is a reliable guide to the next minute.

An important qualification. These figures describe the *average* resting order across everyone who posts one, not the results of professional market makers. Professional firms receive rebates, sit at the front of the queue and can cancel ahead of many adverse moves, so their fills are better than average. A new or small participant at the back of the queue should expect results no better than these.

3 Project Plan

3.1 Changes and decisions since the proposal

- **Spreads are pinned at one tick on all three contracts**, not only on ETH as the proposal expected. Liquidity comparisons will therefore use the size waiting at the best prices, and the three-contract comparison is framed around how much one tick is worth.
- **Volatility has two layers.** We now report both the single-model persistence and the within-day half-life, and the simulator recommendation uses a two-layer model.
- **Volatility regimes** (calm, normal, turbulent, from the past 30 minutes) were built from January–February cutoffs and are used as a covariate.

- **The day-level random effect** came out at zero, so those estimates use ordinary least squares with errors clustered by day, which gives the same point estimates.

3.2 Limitations so far

- The data show only the best bid and ask, not the full order book, and cover one exchange over three months of early 2024.
- All results are from January and February; none has yet been checked on the held-out March data.
- Q1 and Q2 measure the average outcome of fills that happened, not the results of any particular strategy. A participant's own results depend on their fee tier and queue position.
- Results describe associations. Volatility, spreads and trading activity drive one another, so none of them should be read as cause and effect.

4 References

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A Additional figures

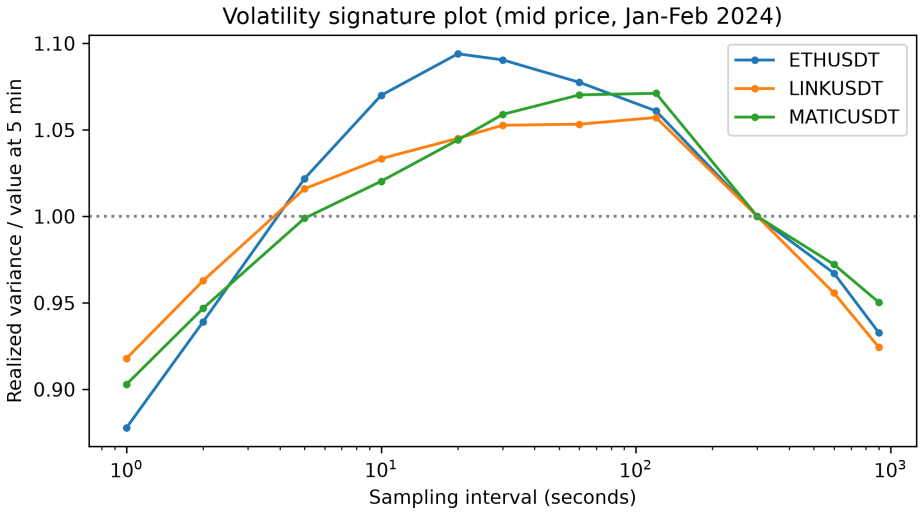


Figure 10: Volatility measured at different sampling intervals, relative to five minutes. The hump shape means price moves tend to continue for a few seconds and partly reverse over minutes. The variation is within about ±10%, small next to the time-of-day effect.

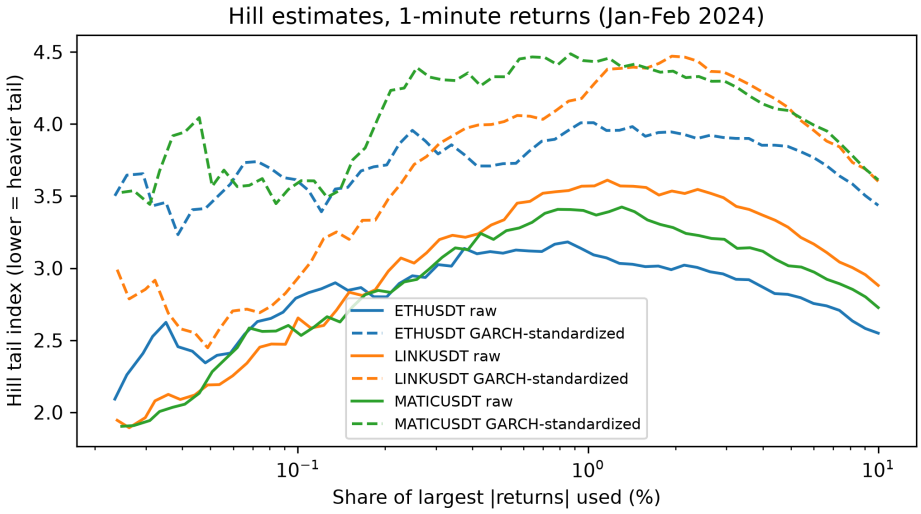


Figure 11: Hill estimates of how heavy the tail of one-minute returns is (lower means heavier), before and after accounting for changing volatility. Raw returns have a tail index of about 3–3.6; after GARCH, about 4–4.4. A normal distribution has no finite tail index.